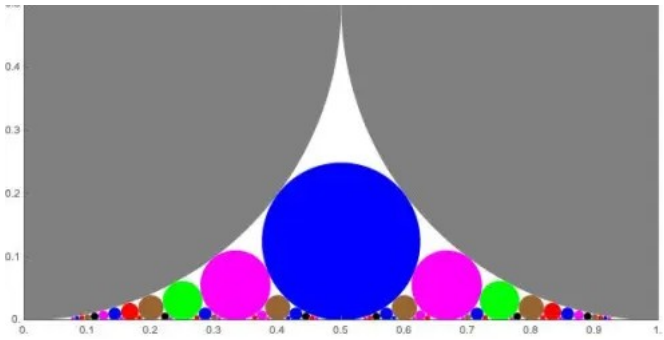
# Ford Circles And Farey Sequences

Jacob Levin

## 1 What are Ford Circles?

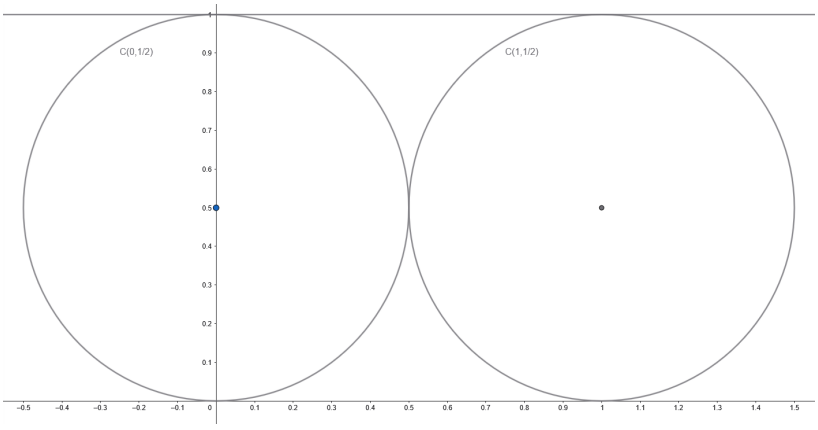
### 1.1 Introduction



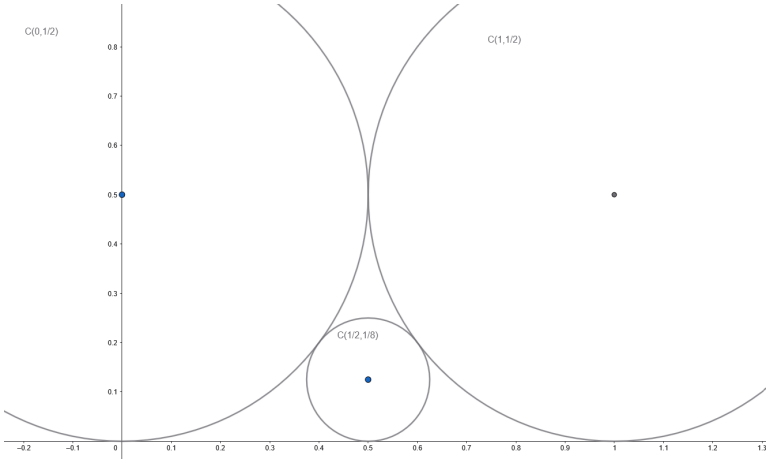
Don't these colorful circles look cool? Everything looks very symmetrical and perfect. What are these mystical, oddly satisfying-to-look-at circles, what rules do they obey, and how do they relate to a seemingly irrelevant sequence of fractions?

### 1.2 Definition

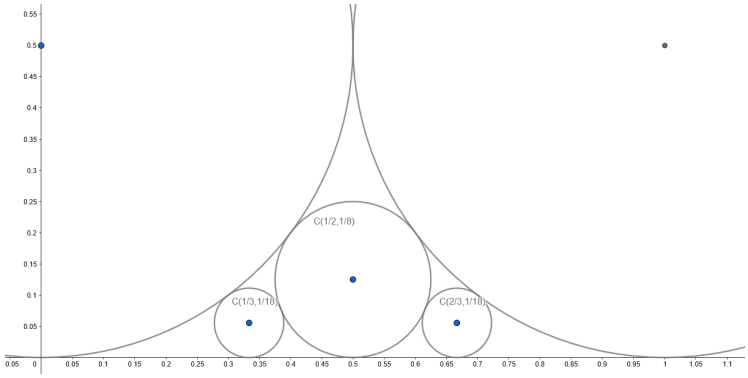
We can say that a circle $C(i, j)$ tangent to the $x$-axis has center $(i, j)$. Next, we realize that the radius of $C(i, j)$ is the $y$-coordinate, or just $j$. Now, consider a set of circles $C$ tangent to the $x$-axis. We are given that $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$ make up the ordered subset $C_1$ (notice that $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$ are tangent to each other, since both have radius $\frac{1}{2}$ and are 1 apart). For $k \geq 1$ $C_{k+1}$ is constructed by starting with $C_k$. Then, for any pair of tangent circles in $C_k$, $C_{k+1}$ contains the $C(i, j)$ tangent to both circles in that pair. For instance, $C_1$ will look like this:




$C_2$ will look like this:




And $C_3$ would look like this:

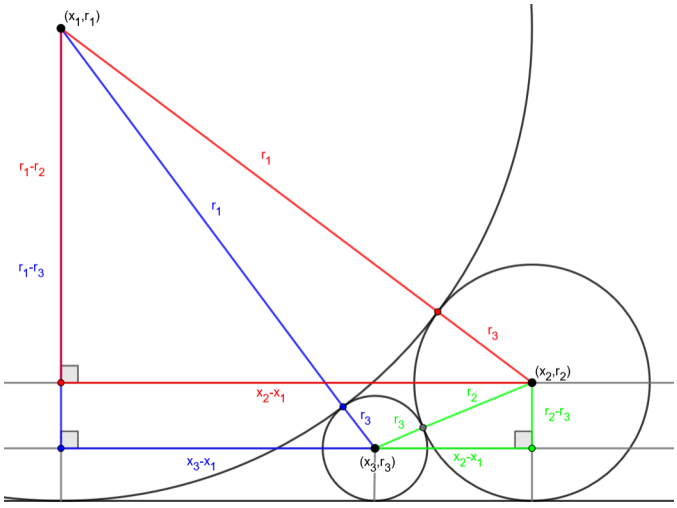



We say that $C$ is the set of all of these circles, or in other words just the union of all $C_k$.

## 1.3 Properties

### 1.3.1 Radius

First, we wish to find the radius of a circle between two others. Suppose that $C(x_1, r_1)$ and $C(x_2, r_2)$ are two tangent Ford Circles. If we say that the Ford circle tangent to them both is $C(x_3, r_3)$ where $r_3 < r_1, r_2$, we want to express $r_3$ in terms of $r_1$ and $r_2$.




As shown in the diagram, we can drop a perpendicular to the lines connecting $C(x_1, r_1)$ and $C(x_2, r_2)$ to their respective tangency points to the $x$-axis. As a result, we get two right triangles, one with hypotenuse $r_1 + r_3$ and legs $r_1 - r_3$ and $x_3 - x_1$, and another with hypotenuse $r_2 + r_3$ and legs $r_2 - r_3$ and $x_2 - x_3$. Applying the Pythagorean Theorem on them both, we get that $(r_1 + r_3)^2 = (r_1 - r_3)^2 + (x_3 - x_1)^2 \implies 4r_1r_3 = (x_3 - x_1)^2 \implies 2\sqrt{r_1r_3} = x_3 - x_1$, and $(r_2 + r_3)^2 = (r_2 - r_3)^2 + (x_2 - x_3)^2 \implies 4r_2r_3 = (x_2 - x_3)^2 \implies 2\sqrt{r_2r_3} = x_2 - x_3$. Adding these gives us $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3} = x_2 - x_1$.

By the same logic, if we construct a similar right triangle with $C_(x_1, r_1)$ and $C_(x_2, r_2)$ and do the Pythagorean Theorem again, we get that $2\sqrt{r_1r_2} = x_2 - x_1$. So since both $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3}$ and $2\sqrt{r_1r_2}$ are equal to $x_2 - x_1$, we have that $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3} = 2\sqrt{r_1r_2} \implies \sqrt{r_3}(\sqrt{r_1} + \sqrt{r_2}) = \sqrt{r_1r_2} \implies \frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$. (Interestingly, this is a degenerate case of Descartes' Circle Theorem, where the $x$-axis actually a circle with infinite radius). So, we have our first property:

**Property 1** For some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(x_1, r_1), C(x_2, r_2)$ such that $r_1, r_2 > r_3$, $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$.

### 1.3.2 Tangency Point

We will make the claim that for some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(\frac{p}{q}, r_1), C(\frac{r}{s}, r_2)$ such that $r_1, r_2 > r_3$ and $\gcd(p, q) = \gcd(r, s) = 1$, $x_3 = \frac{p+r}{q+s}$, and additionally, for some ford circle $C(\frac{a}{b}, r)$ for $\gcd(a, b) = 1$, $r = \frac{1}{2b^2}$. These claims can be proven by induction.

**Base Case:** $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$

The circle tangent to them both will lie directly in the middle by symmetry, since the circles have equal radii and their radial axis is parallel to the x-axis. Thus, the circle tangent to them both touches the x-axis at $x = \frac{0+1}{1+1} = \frac{1}{2}$. By Property 1.1, we also get that $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{\frac{1}{2}}} + \frac{1}{\sqrt{\frac{1}{2}}} \implies \frac{1}{\sqrt{r_3}} = 2\sqrt{2} \implies r_3 = \frac{1}{8} = \frac{1}{2(2^2)}$, so our second condition also holds.

**Induction Hypothesis:** The radii of $C(\frac{p}{q}, r_1)$ and $C(\frac{r}{s}, r_2)$ are $\frac{1}{2q^2}, \frac{1}{2s^2}$, respectively.

**Induction Step:** From our previous right triangles, we have that $2\sqrt{r_1r_3} = x_3 - x_1$ and $2\sqrt{r_2r_3} = x_2 - x_3$. Plugging in the values from the induction hypothesis, we have that $2\sqrt{r_1r_3} = x_3 - \frac{p}{q}$ and $2\sqrt{r_3r_2} = \frac{r}{s} - x_3$. Adding the equations, we get $\frac{r}{s} - \frac{p}{q} = 2\sqrt{r_3}(\sqrt{r_1} + \sqrt{r_2}) \implies \sqrt{r_3} = \frac{\frac{r}{s} - \frac{p}{q}}{2(\sqrt{r_1} + \sqrt{r_2})}$.

Plugging this into $x_3 = \frac{p}{q} + 2\sqrt{r_1r_3}$, we get $x_3 = \frac{p}{q} + 2\sqrt{r_1r_3} = \frac{p}{q} + (\frac{2\sqrt{r_1}(\frac{r}{s} - \frac{p}{q})}{2(\sqrt{r_1} + \sqrt{r_2})}) = \frac{p}{q} + \frac{\sqrt{r_1}(\frac{r}{s} - \frac{p}{q})}{\sqrt{r_1} + \sqrt{r_2}}$.

Using $\sqrt{r_1} = \frac{1}{\sqrt{2}q}$ and $\sqrt{r_2} = \frac{1}{\sqrt{2}q}$, we have that $\frac{\sqrt{r_1}}{\sqrt{r_1}+\sqrt{r_2}} = \frac{\frac{1}{\sqrt{2}q}}{\frac{1}{\sqrt{2}q}+\frac{1}{\sqrt{2}s}} = \frac{\frac{1}{q}}{\frac{1}{q}+\frac{1}{s}} = \frac{s}{q+s}$. This means that $x_3 = \frac{p}{q} + \frac{s(\frac{r}{s}-\frac{p}{q})}{q+s} = \frac{p}{q} + \frac{rq-sp}{q(q+s)} = \frac{pq+ps+rq-sp}{q(q+s)} = \frac{pq+rq}{q(q+s)} = \frac{p+r}{q+s}$, which is our desired $x$-coordinate.

Finally, we want to prove that our third circle has radius $\frac{1}{2x_3^2}$. Since $r_1 = \frac{1}{2q^2}, r_2 = \frac{1}{2s^2}$, and $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$, we have that $\frac{1}{\sqrt{r_3}} = \frac{1}{\frac{1}{\sqrt{2}q}} + \frac{1}{\frac{1}{\sqrt{2}s}} = \sqrt{2}q + \sqrt{2}s \implies r_3 = \frac{1}{(\sqrt{2}(q+s))^2} = \frac{1}{2(q+s)}$. And now, since $x_3 = \frac{p+r}{q+s}$, $r_3 = \frac{1}{2(q+s)^2}$ is our desired result. Thus, we now have two more properties:

**Property 2** For some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(\frac{p}{q}, r_1), C(\frac{r}{s}, r_2)$ such that $r_1, r_2 > r_3$ and $\gcd(p, q) = \gcd(r, s) = 1$, $x_3 = \frac{p+r}{q+s}$.

**Property 3** For some ford circle $C(\frac{a}{b}, r)$ such that $\gcd(a, b) = 1$, $r = \frac{1}{2b^2}$.

# 2 Farey Sequences

Consider the following sequence: for some positive integer $n$, the sequence $F_n$ is the set of reduced fractions in increasing order between 0 and 1 inclusive such that their denominators do not exceed $n$. We can see that $F_1$ is the set $[\frac{0}{1}, \frac{1}{1}]$, $F_2$ is $[\frac{0}{1}, \frac{1}{2}, \frac{1}{1}]$, $F_3$ is $[\frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1}]$, and so on.

## 2.1 Properties

### 2.1.1 $bc - ad = 1$

We will prove that for any two adjacent fractions $\frac{a}{b} < \frac{c}{d}$ in $F_n$, $bc - ad = 1$.

**Proof:** We can show the contrapositive: If two fractions in $F_n$ form a matrix of determinant not $-1$, then they are not adjacent in $F_n$.

Take two Farey fractions $\frac{a}{b} < \frac{c}{d}$ in $F_n$. We have that $bc - ad > 0$. We will suppose that $bc - ad = k > 1$. By Bezout's lemma, there exist integers $x_0, y_0$ such that $bx_0 - ay_0 = 1$. Bezout's lemma also tells us that we can replace $x_0$ and $y_0$ with $x = x_0 + at$ and $y = y_0 + bt$ for any integer $t$ to get the entire solution set to this equation. We take $y$ to be the largest positive integer solution which will satisfy $y \leq n$. Notice that $n - b < y < n$, since otherwise you could find a larger $y$ by adding $b$ and adding $a$ to $x$. Since we defined $y$ to be less than $n$, $\frac{x}{y}$ is in $F_n$. And since $bx - ay = 1$, $bx > ay \implies \frac{x}{y} > \frac{a}{b}$.

Let us examine the difference $\frac{c}{d} - \frac{x}{y} = \frac{cy-dx}{dy}$. Plugging in for $x$, we find that $cy - dx = cy - d(\frac{1+ay}{b}) = \frac{cyb-db-dya}{b} = \frac{y(bc-ad)-d}{b} = \frac{yk-d}{b}$. We wish to prove that $yk - d \geq b$, which would imply that $cy - dx > 0$ and that $\frac{c}{d} > \frac{x}{y}$. Since $bx - ay = 1$, we can treat $(x, y)$ like a vector, so we can write $c$ and $d$ as $c = kx + ma$ and $d = ky + mb$ (we can plug these values into $bc - ad = k$ to see that this is a valid representation). So, we can manipulate $mb = d - ky$. Let us plug in $ky > k(n - b)$ and $d \leq n$: $mb < n - k(n - b) < 2b - n < b$, so $m < 1$. If $m < 0$, then $yk - d = -mb \geq b > 0$, so $cy - dx > 0$. If $m = 0$, then $yk - d = -mb = 0$, however then $cy - dx = 0 \implies \frac{c}{d} = \frac{x}{y}$, which cannot be true (since $bc - ad$ cannot equal 1 and $k$ at the same time). Thus, $yk - d > 0$, so $cy - dx > 0 \implies \frac{c}{d} > \frac{x}{y}$, so $\frac{a}{b} < \frac{x}{y} < \frac{c}{d}$, and we are done.

**Property 4** For any two consecutive fractions $\frac{a}{b}, \frac{c}{d}$ in $F_n$, $bc - ad = 1$.

### 2.1.2 Mediant (Farey Property)

Next, let us look at a fraction in between two fractions: say $\frac{1}{2}$, which is in between $\frac{0}{1}$ and $\frac{1}{1}$. Well, we can see that the numerator of $\frac{1}{2}$ is equal to the sum of the numerators of $\frac{0}{1}$ and $\frac{1}{1}$, and similarly the denominator of $\frac{1}{2}$ is equal to the sum of the denominators of $\frac{0}{1}$ and $\frac{1}{1}$. And surprisingly, as we look at higher cases, we notice that this property also holds for any fraction in $F_n$ (note: the fraction $\frac{a+c}{b+d}$ is known as the mediant of $\frac{a}{b}$ and $\frac{c}{d}$, and will be referred to as such from now on). Let us try to prove that this is the case for all fractions between two others in any Farey Sequence.

**Proof:** We want to show that in $F_k$ all fractions of the form $\frac{x}{y}$ are right between some two fractions $\frac{a}{b} < \frac{c}{d}$ from $F_k$ for whom $\frac{x}{y}$ is the mediant. We know that $bx - ay = 1, yc - xd = 1$ from our previous proof. Next, multiply the first equation by $d$ and the second by $b$, and then sum them. We get $bxd - ayd + ycb - xdb = b + d \implies y(bc - ad) = b + d \implies y = b + d$.

We can conduct a similar process to prove that $x = a + c$, simply by multiplying the first equation by $c$, the second by $a$, and then summing them, to get $bxc - ayc + yca - xda = a + c \implies x(bc - ad) = a + c \implies x = a + c$. Thus, $\frac{x}{y} = \frac{a+c}{b+d}$ and the following property holds:

**Property 5** For any three consecutive fractions $\frac{a}{b}, \frac{x}{y}, \frac{c}{d}$, $\frac{x}{y} = \frac{a+c}{b+d}$.

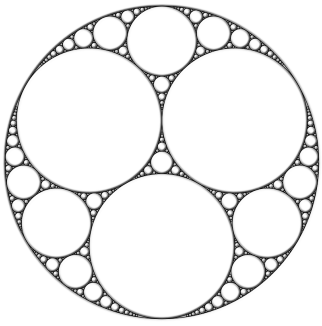
Before we move on, it is important to note that $b + d$ could be greater than $k$. However, our proven property only reveals information about the value of the fraction in between, and since we assumed that all three fractions are in $F_k$, it is implied that the middle fraction could be simplified.

## 2.2 Bringing Everything Together

By Property 2, two tangent circles $C(\frac{a}{b}, \frac{1}{2b^2}), C(\frac{c}{d}, \frac{1}{2d^2})$ are tangent to circle $C(\frac{a+b}{c+d}, \frac{1}{2(c+d)^2})$. Furthermore, by Property 5, for any two consecutive fractions $\frac{a}{b}, \frac{c}{d}$ in a Farey sequence, the fraction in a later sequence which will go between them will be $\frac{a+c}{b+d}$. These properties are very similar, so let us find a way to connect them.

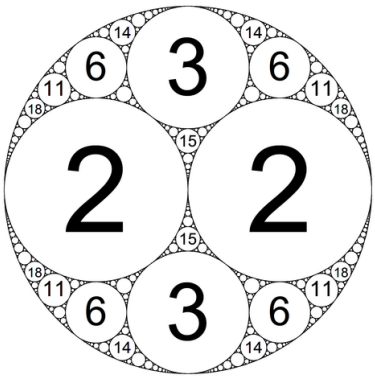
We can prove that the $x$-values of two tangent circles are adjacent in some Farey Sequence, and that the circle tangent to both of them has an $x$-value of the mediant of their $x$-values. Our base case is $C(\frac{0}{1}, \frac{1}{2})$. It is tangent to $C(\frac{1}{1}, \frac{1}{2}), C(\frac{1}{2}, \frac{1}{8}), C(\frac{1}{3}, \frac{1}{18}), ...$, all of which clearly have $x$-values in the Farey Sequence, and are all adjacent to $\frac{0}{1}$ in the Farey Sequence. Next, suppose that some circle $C(\frac{a}{b}, \frac{1}{b^2})$ is tangent to some circles and also adjacent to them in a Farey Sequence. We know that the fractions between $\frac{a}{b}$ and the fractions of those other circles must be their mediants. However, we also know that a circle's $x$-value is the mediant of the $x$-values of the circles tangent to each other and it. So, $C(\frac{a}{b}, \frac{1}{2b^2})$ must be adjacent to that circles as well. Thus, the $x$-values of two tangent circles are adjacent in some Farey Sequence, and that the circle tangent to both of them has an $x$-value of the mediant of their $x$-values.

# 3 Extra: Ford-eque Circles



The image above is of an Apollonian Gasket, a shape created from starting with three circles tangent to each other and indefinitely constructing circles tangent to three others. It is a fun challenge to try to figure out for yourself the radii, if say, the radius of the big circle is 1 and the three circles inside are congruent (here's a hint to start off with: Descartes's Theorem states that for 4 mutually tangent circles, $(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4})^2 = 2(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2})$, however the three circles are internally tangent to the larger one, so the radius of the larger one would have to be considered as $-1$). There exist other Apollonian Gaskets (this term covers a very wide variety of fractals): below is another gasket which has its own properties and

relationships (the numbers inside the circles represent the circle's curvature, or the inverse of the radius, importantly the curvature of the large circle is $-1$), feel free to play around with it as well!




# 4 Citations

thatsmaths. (2023, February 8). Ford-Circles-1. ThatsMaths. https://thatsmaths.com/2023/02/09/ford-circles-farey-series/ford-circles-1/. Accessed 18 October 2025

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